

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. ARCHITECTURAL BLUEPRINT:

The FitBuddy application is constructed utilizing a linear client-server architecture combined with external API integration. This ensures a clean separation of concerns between data collection, backend processing, data persistence, and dynamic presentation.

3. CLIENT-SERVER WORKFLOW & DATA FLOW :

The system architecture operates sequentially through the following phases:

1. **User Input Phase:** The end-user submits their physiological details—specifically their Name, Age, Weight, Goal, and Intensity—via a Jinja2 HTML form.
2. **Backend Interception:** The FastAPI backend route receives the incoming POST request and temporarily stores the submitted data.
3. **AI Orchestration Phase:** The backend logic crafts a specific prompt containing the user's data and transmits it to the Google Gemini API for processing.
4. **Response Synthesis:** The Generative AI model returns a structured 7-day workout plan along with a separate nutrition tip.
5. **Database Persistence:** Utilizing SQLAlchemy, the system commits both the user's profile information and the original AI-generated workout plan directly into the SQLite database for permanent storage.
6. **Interface Rendering:** The final customized fitness plan is passed to the Jinja2 UI, where it is rendered elegantly back to the user on the results page.

4. DATA FLOW SUMMARY :

The overarching data flow of the architecture can be mapped as follows: User UI → FastAPI (routes .py) → Gemini API → SQLite (database .py) → Result UI.

